

CHP-3 Gauss' Law

General Physics 2

2026-09-09

Gauss' Law

Electric Flux

- In previous chapter we already learn how to calculate the electric field of a given charge distribution (whether its discrete or continuous). However that method includes a lot of integration! So in this chapter we will explore a new method for calculating electric fields in **certain** situations.



- In figure $\vec{E}$ is piercing a small square patch with area ΔA . But if you want the components, only E_x goes through area ΔA

Then, amount of electric field passing through the patch ΔA is called **electric flux** $\Delta\Phi$ and defined by

$$\begin{aligned}\Delta\Phi &= E \cos(\theta) \Delta A \\ &= \vec{E} \cdot \Delta\vec{A}\end{aligned}$$

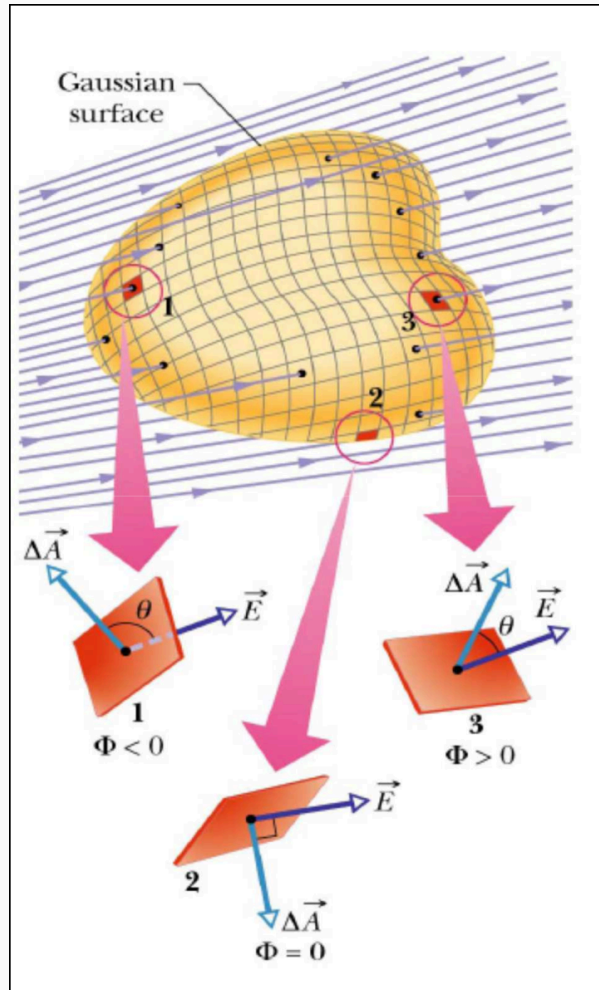
where $\Delta\vec{A} = \Delta A \hat{n}$ with $\hat{n}$ is normal unit vector. In this example it is just $\hat{x}$.

- In order to find the total flux you just need to sum over all surface.

$$\Phi = \sum \vec{E} \cdot \Delta\vec{A},$$

and in the limit $\Delta A \rightarrow 0$ we have the integral representation

$$\Phi = \int \vec{E} \cdot d\vec{A}.$$



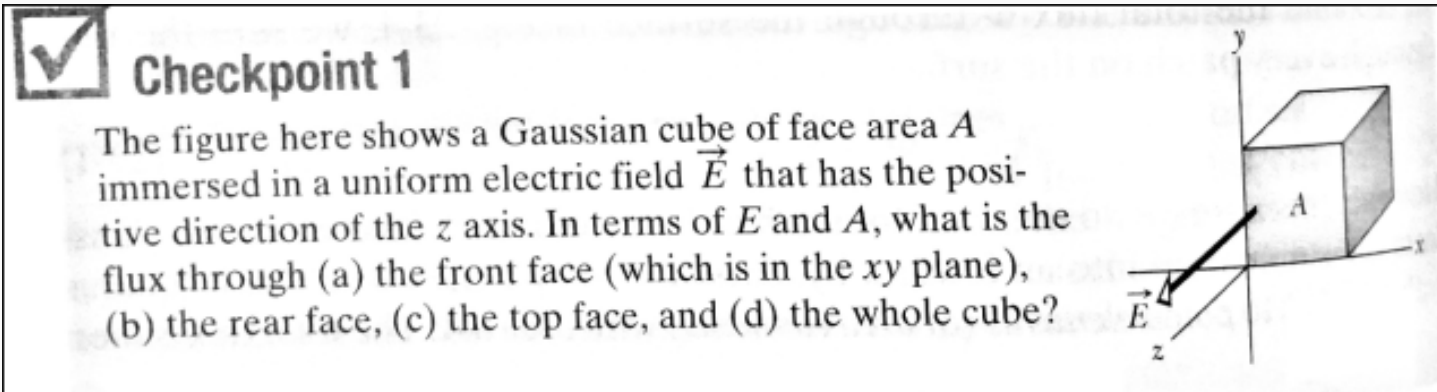
- As seen from figure, the area 1 the normal vector and $\vec{E}$ has $\cos\theta < 0$ as $\theta > \frac{\pi}{2}$ so $\Phi_1 < 0$.
- On 2 $\Phi_3 = 0 = 0$ as $\theta = \frac{\pi}{2}$
- On 3 area $\cos\theta > 0$ as $\theta < \frac{\pi}{2}$ so $\Phi_2 > 0$

From this we can say inward piercing field is (-) flux outward piercing is (+).

- To find total flux through a closed surface we do the integral

$$\Phi = \oint \vec{E} \cdot d\vec{A}$$

- Here small circle on the integral means integration is to be performed over a closed surface. This closed surface is called a **GAUSSIAN SURFACE**.

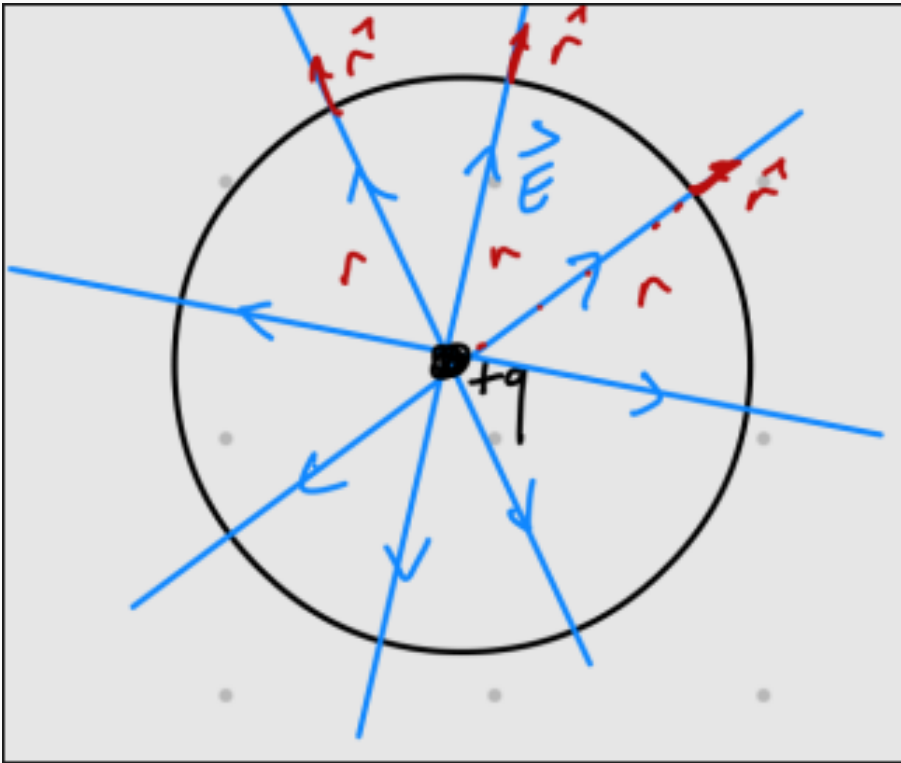


If the volume within a arbitrary closed mathematical surface holds a net electric charge Q then the electric flux through the surface is Q/ϵ_0 .

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

- This law can be used to calculate the electric field provided the charge distribution has a “high degree of symmetry”.

Lets see use this to find the electric field of a point charge. Take the Gaussian surface to coincide with a sphere of radius r . then,



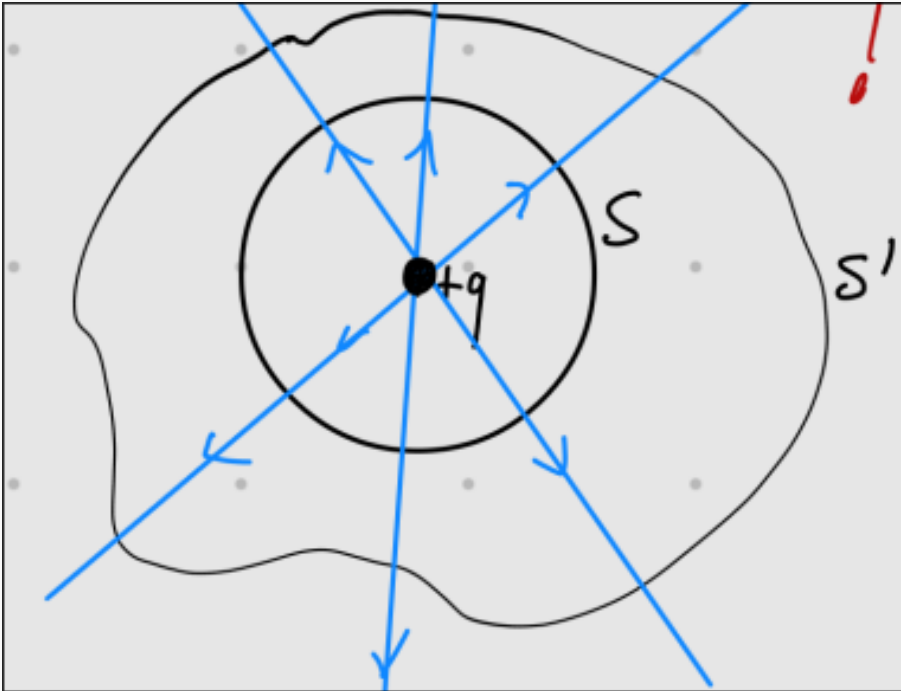
$$\oint \vec{E} \cdot d\vec{A} = Q/\epsilon_0$$

$$E \oint dA = Q/\epsilon_0$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

Note that on the second step we can take E outside of the integral as its **constant** on the integration surface.

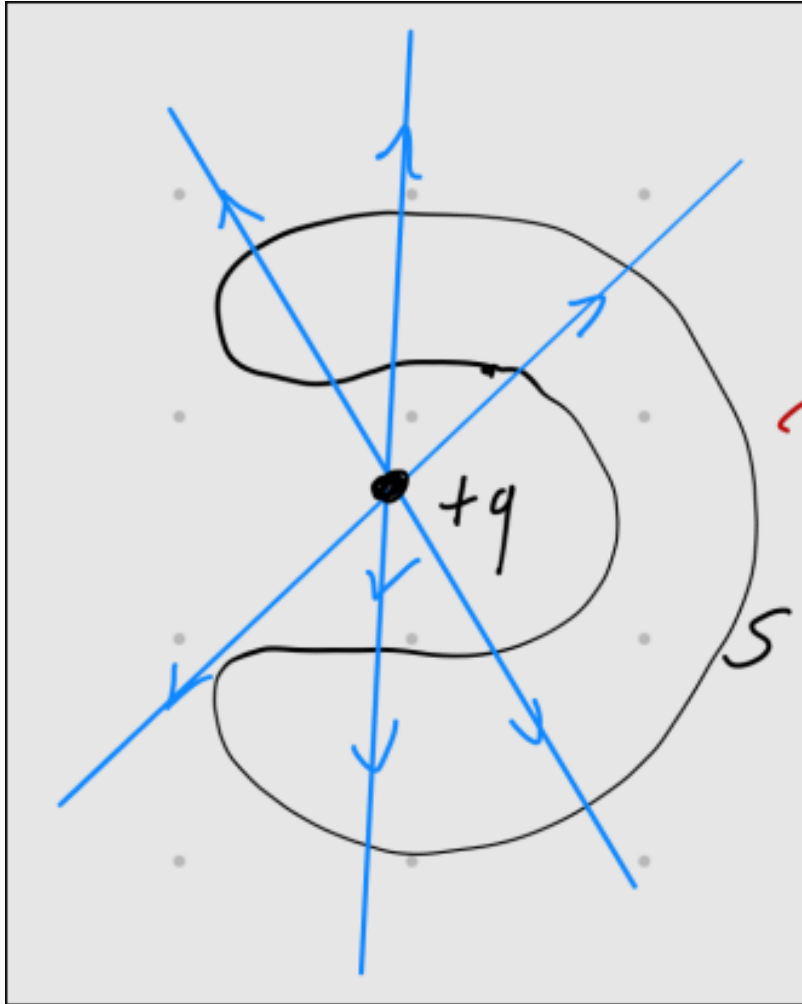
Important: Here we chose the Gaussian surface as a sphere because of symmetry. This choice makes the integration easier. Taking different surfaces that encloses the charge will give same flux!



- The flux through S and S' are the same! So they both satisfy

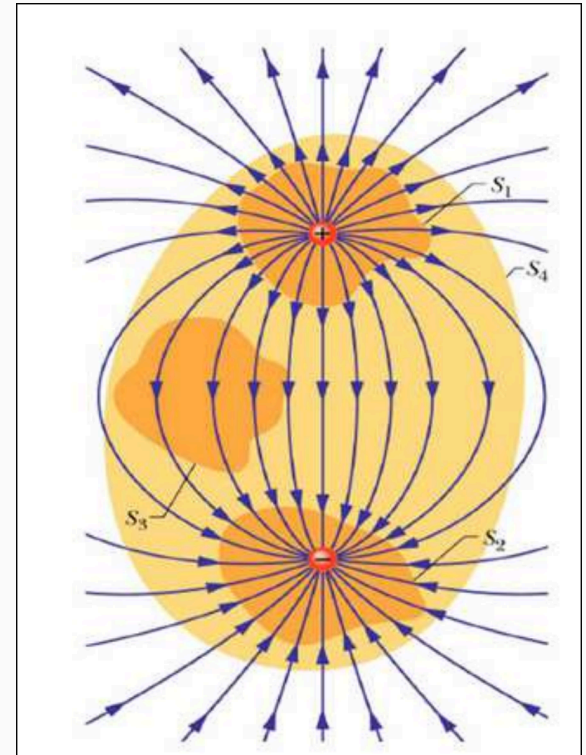
$$\oint \vec{E} \cdot d\vec{A} = Q/\epsilon_0$$

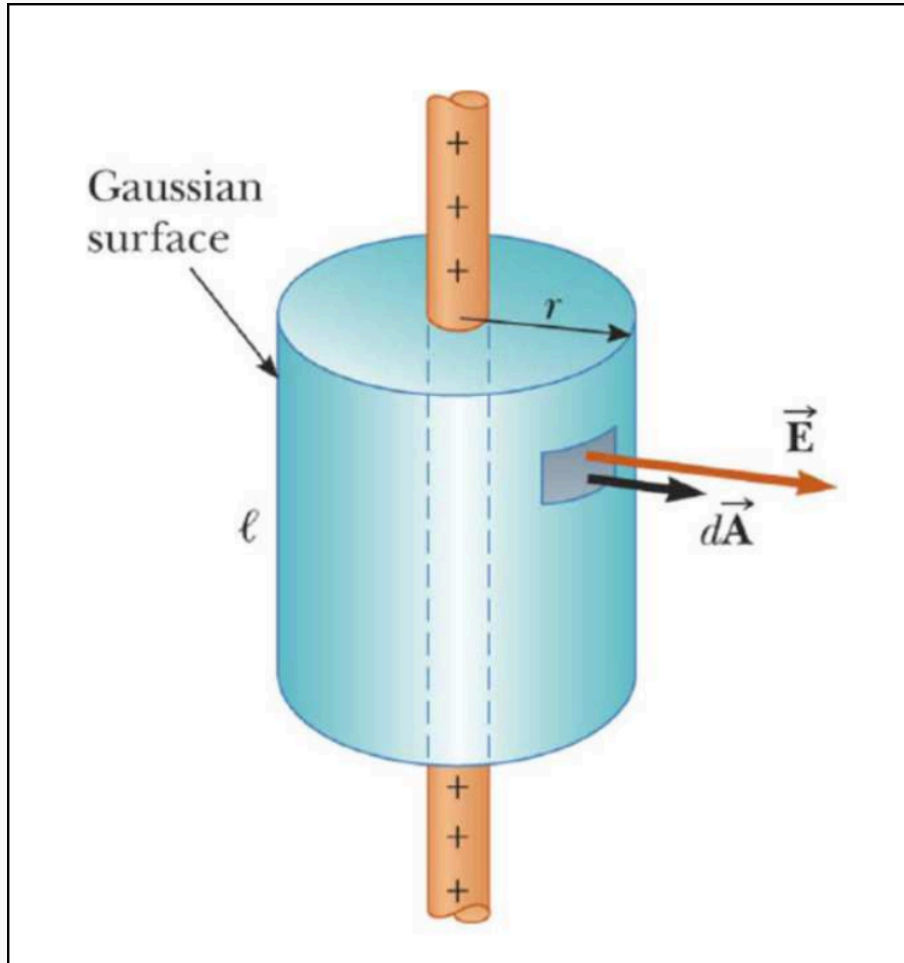
- A fancy way of saying this is integral of flux does not depends on the geometry of the surface, but does depend on topology.



- The flux through S vanishes

$$\oint \vec{E} \cdot d\vec{A} = 0$$





In this example we have a very long thin charged wire with charge density λ . This setup has the cylindrical symmetry,

$$\oint \vec{E} \cdot d\vec{A} = Q/\epsilon_0$$

$$E \oint dA = Q/\epsilon_0$$

$$E(2\pi r h) = \lambda h$$

The final answer is then

$$E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r}.$$

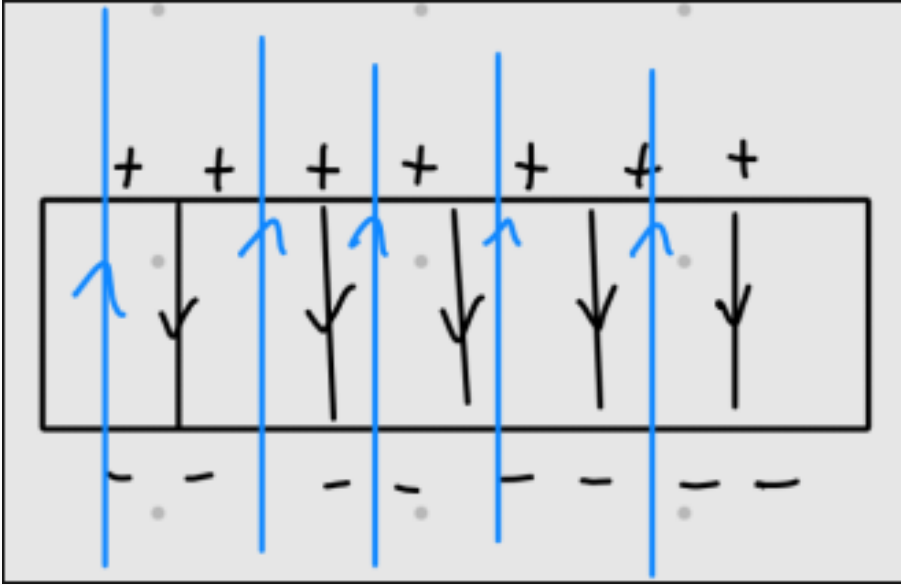
Conductors

Conductors in electric fields

Previous example we computed electric field in a sphere where the charges we consider are distributed homogeneously all over the body. But as we discussed in conductors electrons are mostly free they can move without restraint within the volume of metal. So if we put a conductor in an electric field, the free electrons will accelerate in response.

The electrons will move in the opposite direction to electric field and when the charge distribution on a conductor reaches static equilibrium, the net electric field inside the conductor is exactly zero!

- You can think about the proof of this statement by contradiction: if there was a field inside then free electrons will continue to move

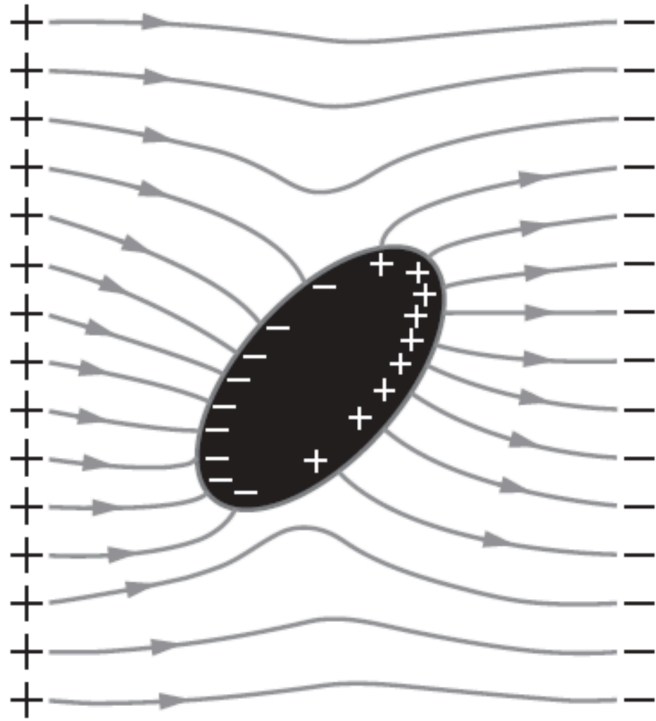


If you immerse a thick conductor slab in electric field (blue lines) the charge that accumulates on surface generate an electric field (black lines) opposite side to itself.

- In a conductor in static equilibrium, any (extra) electric charge deposited on the conductor resides on the face of conductor.

(Proof is by Gauss law: you can think of small gauss surface inside conducting material. Since $\vec{E} = 0$ everywhere, charge also vanish!)



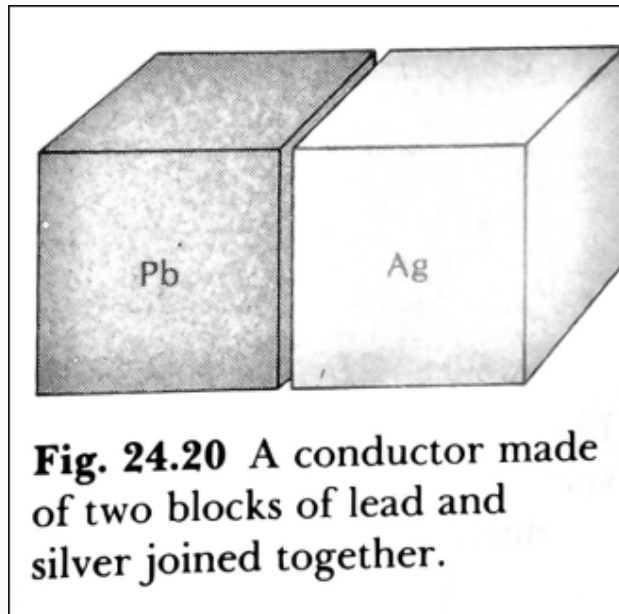


(v) **E is perpendicular to the surface, just outside a conductor.** Otherwise, as in (i), charge will immediately flow around the surface until it kills off the tangential component (Fig. 2.43). (*Perpendicular to the surface, charge cannot flow, of course, since it is confined to the conducting object.*)

The electric field at the surface of a conductor in static equilibrium is perpendicular to the surface.

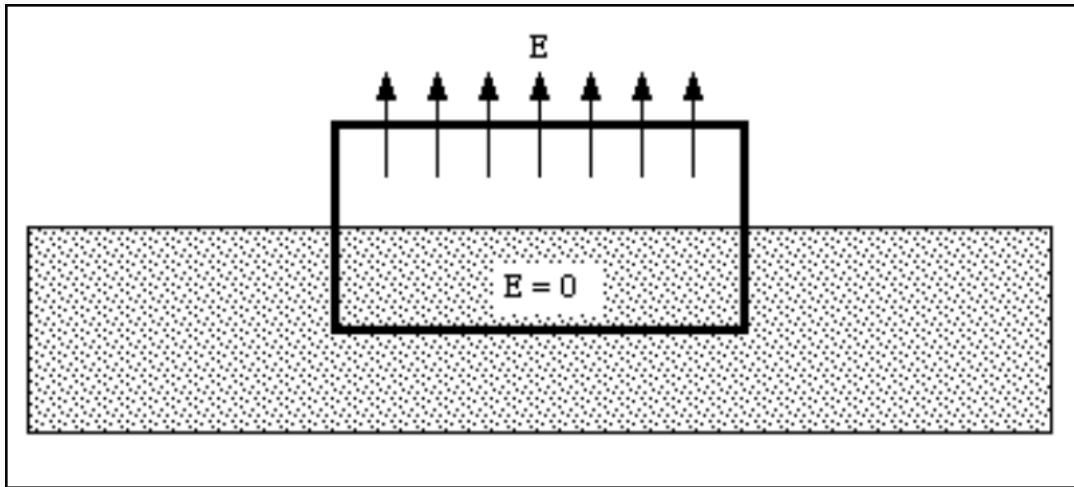
- If the electric field has component tangential to the surface of conductor, the free electrons will move on the surface and there will be no equilibrium.

There was an assumption here about the conductor. We assumed that material is uniform density and uniform chemical composition. For example suppose we have the following setup where a conductor made of two blocks of lead and silver joined together.



- At the interface, an electric field develops even without net charge. Because silver has lower resistivity, electrons flow from lead to silver until their repulsion establishes electrostatic equilibrium.

- We can now find the electric field just outside the conductor surface. Consider a large conductor with the excess positive charge density σ on the surface.



- Since this is a conductor there is no electric field on the face of the cylinder. Only contribution is from the top surface and it is perpendicular to the top surface A .

$$\oint \vec{E} \cdot d\vec{A} = Q/\epsilon_0$$

$$EA = \frac{\sigma A}{\epsilon_0}$$

$$E = \frac{\sigma}{\epsilon_0}$$